Huan Li

Curriculum Vitae

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Ph.D candidate at Department of Computer Science, Zhejiang University, China. Research interests include ***mobile/pervasive computing***, ***augmented reality*** and ***big data management***. Passionate about science, research, with strong technical, business, and interpersonal skills for working in a team and successfully completing a project.

Education

2015–2017 Ph.D Candidate, *Computer Science and Software Engineering*, Zhejiang University, China.
Thesis: "*Research on Spatial Context Awareness and Data Management in Cyber-Physical Systems*" | Details
During the period, I developed my ideas on designing context-dependent applications in large complex indoor environment.

2014–2015 Visiting Student/Research Assistant, *Department of Computer Science*, Aalborg University, Denmark.
I participated in an international academic cooperation project for half a year, working with *Associate Prof.* Hua Lu [8] in the field of indoor big data management.

2012–2014 Ph.D Student, *Computer Science and Software Engineering*, Zhejiang University, China.
GPA – 3.58/4.0 | Detailed List of Exams
I successfully passed the Qualifying Exam (QE) in August, 2014.

2008–2012 Undergraduate, *College of Computer Science*, Sichuan University, China.
I ranked 2 / 466 and got my bachelor degree in June, 2012.

Experience

2015–Present Project, Commercial Indoor Big-Data Analytic Platform for Large Shopping Malls, Database Laboratory, Zhejiang University, China.
This project explored the idea that rapidly-increasing, massive and heterogeneous indoor mobility data can well reflect the behaviors of pedestrians and customers when they hold the mobile devices and walk inside the intelligent sensor-enhanced indoor environments, like a large-scale shopping mall.
Multiple useful and valuable knowledge can be discovered from those captured behaviors, e.g., a business owner of a shop can know the stop duration, (semantic) trajectory and even personal preference of a visitor. It can even benefit more from integrating with other heterogeneous data sources, e.g. the products and sales data collected from the shops, the promotion information collected from the mall, etc.
This project is in cooperation with *Tsinghua University* and a famous high-end shopping mall in Hangzhou downtown. I am one of the two project managers.

2015–Present Research, Literature Repository for Indoor Data Management, Database Laboratory, Zhejiang University, China.

Recently, an increasing number of researches have been conduct on the management of indoor spatial-context data. In the year 2015, I started to build a so-called *Literature Repository* for the researchers or practitioners who are interested in such topics to discuss, to explore, to inspire their thoughts and ideas. A batch of classic literatures in the relevant fields were selected and surveyed in the form of latex-beamer-style slides. Those essential slides and paperworks have been recently published in the GitHub [10] and GitBook [11] synchronously in an open-source style. This ongoing project involves in total 600+ pages PDF, tens of related works from 100+ researchers. In the future, we plan to make it the best site for the indoor big-data fans to learn and to gain.

2015–2016 Research & Project, Toolkit for generating indoor mobility data, Database Laboratory, Zhejiang University, China.

With the proliferation of indoor localization systems, analytics on users' indoor mobility data has emerged as a promising research frontier. However, the database community is missing a synthetic data generator that analysts could use to study algorithms assuming different indoor localization technologies. Such a generator can be used to simulate high volumes of indoor mobility data to evaluate location-dependent queries/algorithms, it can also provide the additional "ground truth" in spite of the discrete nature of the simulated data, which is useful for effectiveness studies in analytics.

We design and implement a generic, open-sourced, user-configurable toolkit for generating different types of indoor mobility data for real-world buildings, the related paper [2] has been accepted by *VLDB'16* – one of the best academic conferences in database field. The detail information of this project can be referred to its web site [3].

2014–2015 Research, Novel Dense-aware Query Processing Techniques in Indoor Environments, Database Group, Aalborg University, Denmark.

During this period, I worked as a visiting researcher (advised by *Associate Prof.* Hua Lu). My research in *Aalborg University* was focused on the novel data management and query processing techniques in complicated indoor environments. We studied on two interesting positioning data uncertainties caused by indoor localization algorithms, namely the spatial uncertainty and temporal uncertainty. Combined with the characteristic of indoor moving objects, we designed efficient and effective query processing algorithms to address various problems which are useful and even important in people's daily life (e.g., congestion control and security management).

2013–2014 Research, Vision-Enhanced Context Computing for Smart Devices, Database Laboratory, Zhejiang University, China.

In this work, my research mainly focused on the design of spatial-context computing framework for smart devices. The main aim is to design a generic and extensible architecture, able to be adapted to heterogeneous environments and able to offer different spatial-context based services to a wide range of devices. In that sense, we worked on the analysis of sensors available on smartphones (Wi-Fi interface, camera and gyroscope), since these are the source of information to identify the place where users are moving around and how their devices are heading to (relative positions & camera poses).

Inspired by SfM (Structure-from-Motion) and SLAM (simultaneous localization and mapping) techniques in computer version, we proposed a vision-enhanced context computing framework. We first use metric learning technique to fuse the features extracted from multiple sensors for quickly searching the covisibility images in the offline database. We then efficiently compute the position and camera pose of device from the structure constructed by a cluster of covisibility images. Our work [1] has been published in *ACM UbiComp'15*, one of the topmost international conferences in computer science.

2012–2013 Research & Project, Cyber-Physical Language & Context-aware Message System, Database Laboratory, Zhejiang University, China.

In this year, I took participation in the *Cyber-Physical System R&D Group*. In this group, we investigated and discussed the ongoing ideas for developing the conception of *Cyber-Physical* and also the implementation of *Augmented Reality* (AR) techniques.

During the period, our group published a work [5] that proposes a novel lightweight XML-based markup language mechanism allowing for organizing a set of related physical objects. To better demonstrate our idea, I undertook the work to build a prototype of context-aware message system for users to immensely send and receive message with perceiving the physical world context. It is worth noting that a hybrid localization module was implemented in this prototype to provide seamless localization service during the navigation of cyber-physical web from outdoor space to indoor space or vice versa, and that is the first time I got to indoor localization.

2012 Project, LBS Music Recommendation System, Database Laboratory, Zhejiang University, China.

In this project, we aimed at developing a *LBS Music Recommendation System* named *"Pictune"* on the basis of a published research paper [4] of our lab. This project is in cooperation with *NetEase*. I worked as a front-end developer. To make a cross-platform production, we used javascript to create apps for iOS, Android and Windows Phone respectively.

2012 Undergraduate Graduation Design, Design of Direction-Aware Tourism Information Search Engine, Database Laboratory, Zhejiang University, China.

This work was supervised by *Prof.* Lidan Shou (Zhejiang University) and *Associate Prof.* Xuwei Li (Sichuan University) when I was visiting the Database Laboratory of Zhejiang University, it's a Java web application for searching tourism information by the input of direction. The client users can view the searched tourism information in a way of augmented reality. I improved the query processing algorithm in a research paper [6] to support my application.

2011 The Innovative Experimental Program, Design and Implementation of AR-based LBS Social Software on Smartphone, Sichuan University, China.

This program was under the supervision of *Associate Prof.* Hui Li, we designed and implemented an augmented reality (AR) browser in android platform, I worked as the back-end and app developer, our application helps users leave message and also multimedia information to each other from a view of 3D real-world. This application was awarded for the 1st Prize in the *Computer Software Contest of Sichuan Province*.

Doctoral Thesis

Title **Research on Spatial Context Awareness and Data Management in Cyber-Physical Systems**

Supervisors *Prof.* Gang Chen and *Prof.* Lidan Shou

Description Cyber-Physical Systems (CPSs) present a deep intertwinement of physical and computational elements, which is highly dependent on the spatial context information obtained from human-interactive smart devices. The computation of such spatial context and the related data management problem have attracted wide attention from many research communities and industries. Aiming at supporting multiple-typed applications based on the spatial context of smart devices, my thesis explores the most advanced theoretical and technical research on spatial context awareness and data management issues. First, in order to obtain a good balance between accuracy, response time and scalability in computing spatial context, I devised a novel multi-modal framework to take advantage of various built-in sensor data, and developed a real-time vision-enhanced system for precisely computing spatial context. Furthermore, to better make use of the massive spatial context data, I also designed efficient spatial-temporal indices and proposed effective processing algorithms to address multiple queries that are practically useful in real-life scenarios.

Scholarships and Distinctions

2016 Alibaba Tianchi Big-Data Algorithm Contest - *spatial-temporal distribution prediction of airport passengers* (top-10 / 3455)

2016 *Award of Honor for Graduate of Zhejiang University* (2015-2016)

2016 *Outstanding Ph.D Student Scholarship of Zhejiang University* (2015-2016)

2016 *VLDB Fellowship Program Winner*

2015 *Triple A Graduate Award of Zhejiang University* (2014-2015)

2015 *Award of Honor for Graduate of Zhejiang University* (2014-2015)

2015 *Outstanding Ph.D Student Scholarship of Zhejiang University* (2014-2015)

2014 *Outstanding Graduate Leader Award of Zhejiang University* (2013-2014)

2013 *Award of Honor for Graduate of Zhejiang University* (2012-2013)

2012 *Creative Talents Award of Sichuan University* (2011-2012)

2011 *China National Scholarship for Undergraduate Students*

2010 *China National Scholarship for Undergraduate Students*

2009 *China National Scholarship for Undergraduate Students*

Social Activities

2016 I was invited to review papers for *TKDE*.

2015 Invited reviewer, *Journal of Mobile Information Systems*.

2015 Invited reviewer, *SIGSPATIAL 2015*.

2014 In charge of visa application and volunteer recruitment, *VLDB 2014* [9].

2013 Campus Ambassador at Zhejiang University, *Hulu Inc*.

2012 Monitor of the *No.1 PhD Class of 2012*, Zhejiang University.

2009 Monitor of the *College's Innovation Experiment Class*, Sichuan University.

Technical Skills

Programming *java, python, C, matlab*

Tools *linux, LaTeX, redis, elk stack, mongodb, docker, hadoop, spark*, etc.

Algorithms have an intimate knowledge of *information retrieval*, *machine learning*, *data mining* and *data visualization*.

Communication Skills

2016.09 *Presentation* at the *VLDB International Conference* in Delhi, India.

2015.09 *Poster Presentation* at the *UbiComp International Conference* in Osaka, Japan.

2015.07 *Oral Presentation* at the *Sensor-enhanced Social Media (Sesame) Workshop* [7], held at Singapore.

Languages

Chinese **mother tongue**

English **intermediate** *Conversationally fluent*

List of Exams

Exam	Grade	Credit
Theory of Computing	94	2
Advanced Operation System	94	2
Advanced Computer System Architecture	90	2
Advanced Computer Network	93	2
Applied Mathematics for Computer Science I	90	2
Applied Mathematics for Computer Science II	92	2
Introduction to Artificial Intelligence	89	2
Data Mining	86	2
Introduction to Machine Learning	87	2
Network Multimedia Search Engine	89	2
	Total	20
	Gpa	**3.58**

Interests

Reading, Writing, Travel, Cooking, Soccer, Violin.

Publication

[1] **Huan Li**, Pai Peng, Hua Lu, Lidan Shou, Ke Chen, Gang Chen. E^2C^2: efficient and effective camera calibration in indoor environments. in *UbiComp*, 9–12, 2015, ACM.

[2] **Huan Li**, Hua Lu, Xin Chen, Gang Chen, Ke Chen, Lidan Shou. Vita: A Versatile Toolkit for Generating Indoor Mobility Data for Real-World Buildings. in *PVLDB* 9(13):1453–1456, 2016, ACM.

Reference

[3] Vita Project. http://db.zju.edu.cn/vita/.

[4] Ke Chen, Gang Chen, Lidan Shou, Fei Xia. Pictune: situational music recommendation from geotagged pictures. in *SIGIR*, 1011–1011, 2012, ACM.

[5] Xiaoling Gu, Lidan Shou, Hua Lu, Gang Chen. A generic framework for cyber-physical web. in *Middleware*, 1, 2013, ACM.

[6] Guoliang Li, Jianhua Feng, Jing Xu. Desks: Direction-aware spatial keyword search. in *ICDE*, 474–485, 2012, IEEE.

[7] SeSaMe workshop 2015. http://sesame.comp.nus.edu.sg/2015workshop/.

[8] Hua LU's homepage. http://people.cs.aau.dk/~luhua/.

[9] VLDB2014 - Acknowledgement. http://www.vldb.org/2014/acknowledge.html.

[10] Literature Repository for Indoor Data Management. https://longaspire.github.io/intro2idm-gh/.

[11] Indoor Data Management, GitBook. https://longaspire.gitbooks.io/indoor-data-management/content/.

Paper in Submission

[a] **Huan Li**, Hua Lu, Lidan Shou, Gang Chen, Ke Chen. Flow Analysis on Uncertain Historical Indoor Mobility Data. to *ICDE*, 12 pages.

[b] Hua Lu, **Huan Li**, Lidan Shou, Gang Chen, Ke Chen. In Search of Indoor Dense Regions: An Approach Using Indoor Positioning Data. to *ICDE*, 12 pages.